

Delayed Visual Feedback and the Sense of Agency:

A Budget-Margin Protocol

Study protocol, version 1.0

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Abstract

We test whether an acute loss of the sense of agency — the laboratory analogue of “not feeling like myself” — behaves like a *complexity-budget overrun*. Participants trace a ring with a mouse while a visible dot follows the hand with a lag of 0–300 ms. The lag manufactures a conflict between recent motor history and present visual feedback (a “multiply-connected communicated past” in the lab’s framework), which the budget-margin account predicts should (H1) reduce rated agency, (H2) raise computable complexity shadows of the tracing behaviour, and (H3) do so in a coupled way, with the shadow tracking agency within participants after tracing error is controlled. An attention-control condition rules out surprise. A planned second stage tests whether an independently measured capacity index (a proxy for OCOS lateral inhibition β) moderates susceptibility (H4). The design is within-subjects, single-session, browser-based (jsPsych), $N = 30$, and fully specified below so that the analysis can be run as pre-registered.

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1 Background and rationale

The everyday report “I don’t feel like myself today” most often describes a disturbance of the *sense of agency* — the feeling of authoring one’s own actions — rather than of body ownership (Gallagher, 2000; Haggard, 2017). On comparator accounts, agency reads out the match between predicted and actual sensory consequences of action (Blakemore et al., 2000; Synofzik et al., 2008); delaying visual feedback of one’s own movement is the standard way to break that match.

This protocol re-describes the same manipulation in the terms of the lab’s complexity framework (Sengupta, in prep.f). The self-summary the brain maintains is treated as a bounded-capacity sufficient statistic: it can jointly track at most K interacting subsystems (the *budget*), while a situation demands the joint tracking of d_{required} subsystems, a quantity set by the coupling structure of the currently active system (formally treewidth + 1). The *budget margin* is $m = K - d_{\text{required}}$; feeling like oneself is $m \geq 0$, and acute self-alienation is $m < 0$. A delayed feedback channel adds a second, time-shifted pathway from motor output to visual input — a multiply-connected communicated past — which introduces cycles into the active coupling graph and raises d_{required} . The account therefore predicts not just that agency falls with delay, but that a *complexity* signature of the behaviour rises with it, and that the two move together. The lab’s on-centre off-surround (OCOS) work identifies lateral-inhibition strength β as the mechanism that sets K across numerosity and working memory (Sengupta et al., 2014, 2017; Verma & Sengupta, 2023; Sengupta, 2025), yielding the stage-2 moderation prediction.

Because treewidth cannot be measured from behaviour, the framework’s constructs are operationalised by computable *shadows*: sample entropy of hand speed and cross-recurrence determinism between hand and displayed cursor (Marwan et al., 2007), chosen because dimensionality and recurrence measures are established readouts of how organised a system’s activity is (Mazucato et al., 2016).

2 Hypotheses

Pre-registered predictions

H1 (agency). Rated agency decreases monotonically with feedback delay, with the steepest change between 60 and 180 ms.

H2 (shadow). Complexity shadows of tracing — sample entropy of hand speed and the loss of hand–cursor cross-recurrence determinism — increase with delay, *after controlling for radial tracing error*.

H3 (coupling). Within participants, a stretch’s shadow value predicts its agency rating after delay and error are controlled. This is the distinctive budget-margin prediction: the shadow is not merely a second effect of delay but tracks the felt loss of agency.

H4 (stage 2, moderation). An independently measured capacity index (proxy for β) moderates the agency–delay slope: lower capacity, steeper decline and earlier shadow rise.

Control prediction. An equally surprising but non-self-caused event (a brief colour flash of the cursor at 0 ms) produces no reliable change in agency or shadows relative to plain 0-ms stretches.

What would count against the account. H1 is a replication of a well-established effect and its failure would indicate a task problem, not a theoretical one. The account is falsified in its distinctive part if H2 holds but H3 fails (delay raises complexity and lowers agency *independently*), or if the flash control produces effects of the same size as delay.

3 Design

Within-subjects, single session, one factor (feedback delay) with five levels and one control condition.

Condition	Delay (ms)	Stretches
D0	0	6
D60	60	6
D120	120	6
D180	180	6
D300	300	6
Flash control	0, with a 300-ms red flash of the cursor	6
Total main		36
Practice	0, 300, 120	3

Each stretch lasts 20 s of continuous tracing followed by a single agency rating. The 36 main stretches are fully randomised per participant and split into three blocks of 12 with self-paced breaks. Session length is approximately 25 min (tracing $\approx$ 13 min).

Note

Delay is fixed *within* a stretch and varies only *between* stretches. This keeps each rating attached to one well-defined delay and lets the complexity shadows be computed on a stationary segment. A continuously jittered delay is a natural follow-up but is not part of this protocol.

4 Participants

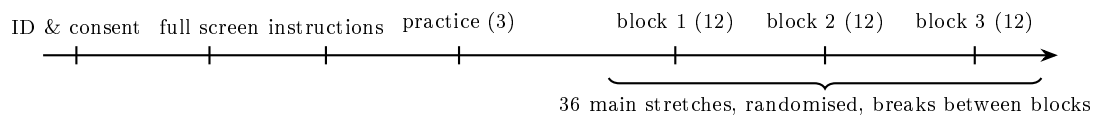
- Target $N = 30$ analysable participants (adult volunteers from the Krea community, 18–40 years), recruited after a 5-participant technical pilot whose data are not included.
- Inclusion: normal or corrected-to-normal vision; right-handed mouse use preferred but not required (handedness recorded).
- Exclusion (a priori): self-reported neurological or motor condition affecting hand movement; use of a trackpad instead of a mouse.
- Informed consent obtained before the session; participation is voluntary, no deception beyond not announcing the delays in advance; full debrief afterwards.

Sample-size rationale. Delay effects on rated agency are large in the literature; the limiting hypothesis is H3, a within-participant coupling detected as a fixed effect in a mixed model with ≈ 30 stretches per participant. With $N = 30$ and 30 non-control stretches each ($n = 900$ observations), a standardised within-person slope of 0.2 is detectable with power > 0.8 under conventional assumptions about the intra-class correlation; this is a planning estimate, and the pilot will be used to refine it before data collection begins.

5 Apparatus

- Implementation: jsPsych 7, single HTML file (`index.html`), run in Chrome or Firefox in full-screen mode on a laboratory desktop. A wired optical mouse; the trackpad is disabled.
- Display: 60 Hz or higher; the canvas is 600×600 px, ring radius 180 px, cursor dot radius 8 px. The operating-system pointer is hidden over the canvas so that only the lagged dot is visible.
- Timing: hand position is sampled on every `mousemove` event with `performance.now()` timestamps; the dot is drawn on every animation frame at the interpolated hand position from $t - \Delta$. Logged frame interval and sample count are saved per stretch.
- Intrinsic latency: the mouse-to-photon latency of the setup is measured once with a high-speed camera (or a photodiode against a known input) and reported. All stated delays are *added* to this constant; it does not vary across conditions and so cannot confound them.

6 Task and procedure



1. The participant enters an ID; the experiment enters full screen.

2. Instructions: trace around the grey ring smoothly and continuously at a steady, comfortable pace; a blue dot follows the hand and may sometimes lag; keep tracing regardless; after each go, answer one question.
3. **One stretch.** An overlay reads “Move the mouse onto the ring to begin.” The clock starts on the first movement inside the canvas. For 20 s the participant traces while the dot is displayed at the hand position from $t - \Delta$. A small countdown is shown in the corner. In flash-control stretches the dot turns red for 300 ms at a random moment between 5 and 15 s.
4. **Rating.** A slider (0–100) with the prompt “How much did it feel like *you* were moving the dot?” and end-labels “Not at all me” / “Completely me”; the slider must be moved before continuing.
5. Three practice stretches (0, 300, 120 ms), then 36 main stretches in three blocks with breaks.
6. Debrief question (what, if anything, was noticed about the dot), then the data file downloads automatically.

7 Measures

7.1 Primary dependent variable

Rated agency: the slider response (0–100) after each stretch.

7.2 Complexity shadows (secondary DVs)

Computed offline (`analyze_tracing.py`) on trajectories linearly resampled to a uniform 50 Hz grid, so that frame-rate jitter cannot masquerade as complexity.

- **Sample entropy of hand speed** ($m = 2$, $r = 0.2SD$): higher values indicate less regular tracing.
- **Cross-recurrence determinism** between the hand (x, y) and dot (x, y) trajectories (radius = 10% of the maximum cross-distance, minimum diagonal line length 2, trajectories subsampled to ≤ 900 points): lower determinism indicates the two have decoupled. Recurrence rate is stored alongside as a check.

7.3 Covariates and checks

- **Radial tracing error:** mean absolute distance of the *hand* from the ring (px). Entered as a covariate in every model.
- **Revolutions** per stretch (pacing check) and mean frame interval / sample count (timing quality).
- **Debrief response** (noticed lag / flash / both / nothing).

7.4 Stage 2: capacity index

Before the tracing task, a 5-minute capacity task from the lab’s existing toolkit (visual working-memory change detection or a numerosity task) yields a per-participant capacity estimate used as the β proxy for H4. Its scoring follows the lab’s published procedure and is fixed before H4 data are examined.

8 Data logging

Per stretch the experiment records: condition (`delay_ms`, `flash`, `flash_at_ms`), phase and block, sample count, mean frame interval, mean radial error, revolutions, and the full trajectories `t`, `hx`, `hy`, `cx`, `cy` (hand and displayed dot, at frame rate). The rating trial records the slider value and its response time.

The experiment is hosted on GitHub Pages and posts its data to a Google Apps Script web app that writes one JSON file per session (`agency_tracing_v1_<ID>_<timestamp>.json`, full data) plus a trajectory-free CSV summary into a private lab Drive folder, and appends a line to a save log. Data are posted at every block break and at the end (overwriting the same files), so an interrupted session still leaves a partial file. If the upload cannot be confirmed, the same files are downloaded to the session computer and the completion screen instructs the participant to inform the researcher. A `?save=off` switch exists for technical testing only and is never used with participants.

9 Analysis plan

All confirmatory analyses use linear mixed models with a random intercept per participant; random slopes for delay are added if the model converges and improves fit. Practice stretches are excluded. Unless stated, models are fit on main, non-flash stretches.

- **H1.** $\text{agency} \sim \text{delay} + \text{error} + (1|\text{pid})$. Prediction: negative delay coefficient. Delay is also entered as a factor to test the 60–180 ms contrast.
- **H2.** For each shadow s : $s \sim \text{delay} + \text{error} + (1|\text{pid})$. Prediction: positive slope for sample entropy, negative for determinism, with error in the model.
- **H3.** $\text{agency} \sim s + \text{delay} + \text{error} + (1|\text{pid})$. Prediction: the shadow coefficient is non-zero (negative for sample entropy, positive for determinism) with delay and error controlled. Shadows are within-participant centred so the coefficient reflects within-person coupling.
- **Control.** $\text{agency} \sim \text{condition} + \text{error} + (1|\text{pid})$ on D0 vs. flash stretches; prediction: no reliable difference; the same for each shadow.
- **H4 (stage 2).** Add the capacity index and its interaction with delay to the H1 and H2 models; prediction: negative capacity $\times$ delay interaction on agency.

Effect sizes are reported as standardised coefficients with 95% CIs. Two shadows are tested for H2/H3; a Holm correction is applied within each hypothesis. Exploratory analyses (e.g. order/fatigue effects, agency response times) are labelled as such.

Exclusion and stopping rules

- A stretch is excluded if its sample count implies a mean frame interval > 25 ms, if fewer than 3 revolutions were completed, or if radial error exceeds the participant's mean by more than 3 SD.
- A participant is excluded if more than 25% of stretches are excluded, if a trackpad was used, or if the debrief indicates the instructions were not followed.
- Data collection stops at $N = 30$ analysable participants; no interim significance tests are run.

10 Confounds and controls

Rival explanation	Why it matters	How it is handled
Tracing error	Delay makes tracing worse; error has its own behavioural and felt signatures.	Radial error logged continuously and entered as a covariate in every model.
Surprise / attention	A sudden lag grabs attention; effects could be distraction.	Flash-control stretches: equally salient, not self-caused. Predicted null.
Dual-task load	Rating while tracing changes the tracing.	Rating occurs only <i>after</i> each stretch.
Learning and fatigue	Agency and precision drift over 36 stretches.	Full randomisation of conditions; block entered as an exploratory covariate.
Demand characteristics	Participants may infer the manipulation and rate accordingly.	Delays not announced; debrief records what was noticed; H3 (a coupling with an implicit behavioural measure) does not depend on ratings alone.

11 Ethics and data management

The study involves a low-risk computer task with no deception beyond withholding the delay schedule; approval is sought from the institutional ethics review before piloting. Data are pseudonymous (participant ID only), stored on lab machines and the lab’s secure drive, and shared on request in de-identified form with the analysis code. Participants may withdraw at any point without penalty.

12 Timeline

Weeks	Milestone
1–2	Ethics submission; measure setup latency; run <code>index.html?dev=1</code> end to end; confirm logging.
3	Technical pilot ($N = 5$); check timing quality, pacing, rating distributions; freeze parameters.
4–7	Main data collection ($N = 30$).
8–9	Pre-registered analyses (H1–H3, control); one figure: agency and each shadow vs. delay.
10	Stage-2 decision: add the capacity task and test H4 if resources allow.
11–12	Write-up: question, method, results, limitations, next steps.

A Parameter table

Parameter	Value	Where set
Delay levels	0, 60, 120, 180, 300 ms	CONFIG.delays_ms
Stretches per delay	6	CONFIG.reps_per_delay
Flash-control stretches	6 (at 0 ms)	CONFIG.n_flash_control
Stretch duration	20 000 ms	CONFIG.stretch_ms
Blocks	3	CONFIG.n_blocks
Canvas / ring radius / dot radius	600 / 180 / 8 px	CONFIG
Flash duration and window	300 ms, 5–15 s	CONFIG.flash_ms, flash_window
Rating scale	0–100 slider, start 50	CONFIG.slider_*
Resampling for shadows	50 Hz	analyze_tracing.py
Sample entropy	$m = 2$, $r = 0.2$ SD	analyze_tracing.py
CRQA	radius 10% max distance, min line 2, ≤ 900 pts	analyze_tracing.py

B Session checklist

- ☐ Mouse connected, trackpad disabled, display refresh checked.
- ☐ Browser full screen; other windows and notifications closed.
- ☐ Consent signed; handedness and any relevant conditions recorded on the session sheet.
- ☐ Participant ID entered exactly as on the session sheet.
- ☐ After the session: confirm `tracing_<ID>.json` downloaded and opens; copy to the lab drive.
- ☐ Note anything unusual (interruptions, pauses, comments) on the session sheet.

References

- Blakemore, S.-J., Wolpert, D., & Frith, C. (2000). Why can't you tickle yourself? *NeuroReport*, 11(11), R11–R16. <https://doi.org/10.1097/00001756-200008030-00002>
- Gallagher, S. (2000). Philosophical conceptions of the self: Implications for cognitive science. *Trends in Cognitive Sciences*, 4(1), 14–21. [https://doi.org/10.1016/S1364-6613\(99\)01417-5](https://doi.org/10.1016/S1364-6613(99)01417-5)
- Haggard, P. (2017). Sense of agency in the human brain. *Nature Reviews Neuroscience*, 18(4), 196–207. <https://doi.org/10.1038/nrn.2017.14>
- Marwan, N., Romano, M. C., Thiel, M., & Kurths, J. (2007). Recurrence plots for the analysis of complex systems. *Physics Reports*, 438(5–6), 237–329. <https://doi.org/10.1016/j.physrep.2006.11.001>
- Mazzucato, L., Fontanini, A., & La Camera, G. (2016). Stimuli reduce the dimensionality of cortical activity. *Frontiers in Systems Neuroscience*, 10, 11. <https://doi.org/10.3389/fnsys.2016.00011>
- Sengupta, R., Surampudi, B. R., & Melcher, D. (2014). A visual sense of number emerges from the dynamics of a recurrent on-center off-surround neural network. *Brain Research*, 1582, 114–124.
- Sengupta, R., Bapiraju, S., & Melcher, D. (2017). Big and small numbers: Empirical support for a single, flexible mechanism for numerosity perception. *Attention, Perception, & Psychophysics*, 79(1), 253–266.

- Sengupta, R. (2025). Modeling visual working memory using recurrent on-center off-surround neural network with distance dependent inhibition. *Scientific Reports*, 15, 23000. <https://doi.org/10.1038/s41598-025-06919-5>
- Sengupta, R. (in preparation). *The Felt Machine: Treewidth, communicated pasts, and the computational body schema*. Lab manuscript, Computational Cognition Laboratory, Krea University.
- Sengupta, R. (forthcoming). *Mind, Model, and Complexity*. River Publishers.
- Synofzik, M., Vosgerau, G., & Newen, A. (2008). Beyond the comparator model: A multifactorial two-step account of agency. *Consciousness and Cognition*, 17(1), 219–239. <https://doi.org/10.1016/j.concog.2007.03.010>
- Verma, B. K., & Sengupta, R. (2023). Emergence of behavioral phenomena and adaptation effects in human numerosity decoder using recurrent neural networks. *Scientific Reports*, 13, 19352. <https://doi.org/10.1038/s41598-023-44535-3>